

محاكاة إطلاق صاروخ فضائي Space Rocket Lunch Simulation

مشروع مادة الحسابات العلمية



الدراسة الفيزيائية والقوانين:

القوى المؤثرة:

Lift:

$$L = \frac{1}{2} \cdot C_l \cdot \rho \cdot A \cdot v^2$$

Drag:

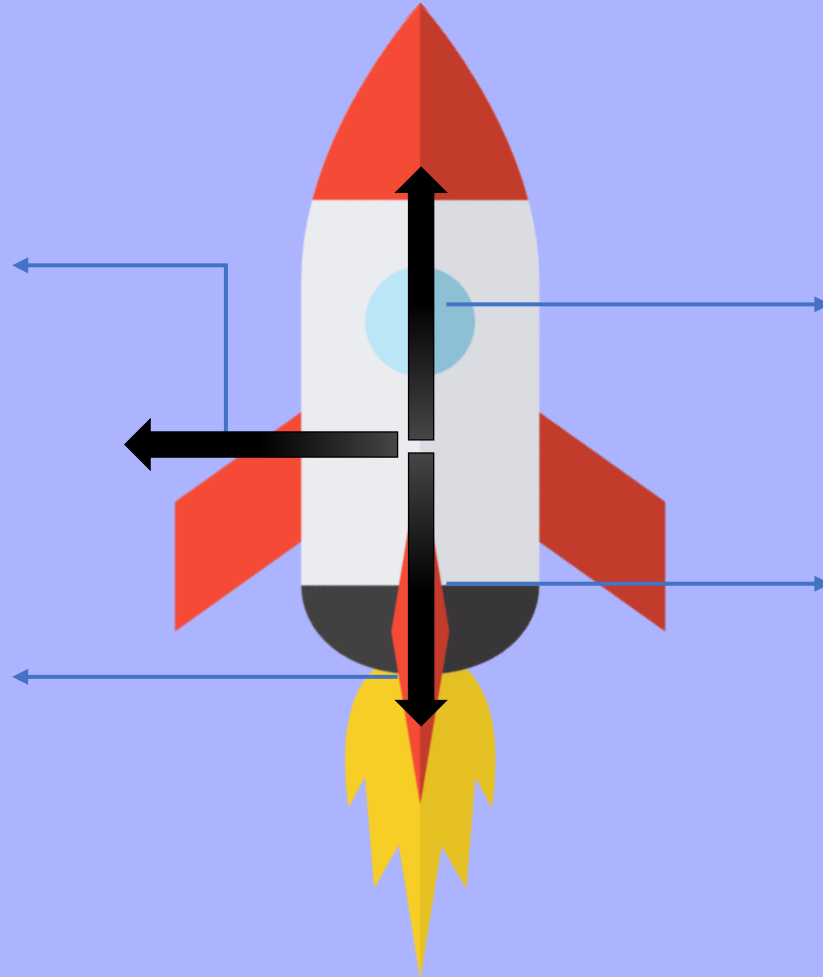
$$D = \frac{1}{2} \cdot C_d \cdot \rho \cdot A \cdot v^2$$

Thrust:

$$T = v_e \cdot \frac{dm}{dt} + (P_c - P_0) \cdot A_e$$

Weight:

$$W = \frac{m_1 \cdot m_2 \cdot G}{r^2}$$



الدراسة الفيزيائية والقوانين:

السرعة الحدية أثناء السقوط:

$$v_t = \sqrt{\frac{2 \cdot m \cdot g}{\rho \cdot A \cdot C_d}}$$

```
function terminalVelocity() {  
  let A = Math.PI * this.rocketRadius * this.rocketRadius;  
  let rho = this.air_rho();  
  
  return Math.sqrt((2 * this.rocketMass * this.gravity) / (rho * A * this.CD));  
}
```



الدراسة الفيزيائية والقوانين:

السرعة الحدية أثناء الصعود:

$$v_r = c \cdot \tanh \left(\frac{I_{sp}}{c} \cdot \ln \left(\frac{m_0}{m_1} \right) \right)$$

```
function maximumVelocity() {  
  let r = this.specificImpulse / World.speedOfLight *  
    Math.log(fMass + this.rocketMass) / this.rocketMass;  
  return World.speedOfLight * 10 * Math.tanh(r);  
}
```





قوة الدفع:

```
function thrust(dTime) {  
    // F = massFlowRate * Ve + (Pc - P0) * Ae  
    let Ae = Math.PI * this.nozzleRadius * this.nozzleRadius;  
    let P0 = this.atm_pressure();  
    let tx = (this.massFlowRate * dTime *  
        this.exhaust_velocity() +  
        (this.chamberPressure - P0) *  
        Ae) *  
        Number(  
            Math.cos(  
                this.rocketAngle *  
                Math.PI / 180  
            ).toFixed(10)  
        );  
    let ty = (this.massFlowRate * dTime *  
        this.exhaust_velocity() +  
        (this.chamberPressure - P0) *  
        Ae) *  
        Number(  
            Math.sin(  
                this.rocketAngle *  
                Math.PI / 180  
            ).toFixed(10)  
        );  
    return vector.create(  
        tx, ty  
    );  
}
```



قوة الثقل:

```
gravity_acceleration() {  
    let instantaneousHeight = this.position.getY() + World.earthRadius;  
    let d = instantaneousHeight * instantaneousHeight;  
    this.gravity = (World.GravitationalConstant * World.earthMass) / d;  
}  
  
gravity_force() {  
    return vector.create(0, -1 * (this.fuelMass + this.rocketMass) * this.gravity);  
}
```



قوة السحب:

```
function drag() {  
  // R = 1/2 * Cd * rho * A * V^2  
  let velocitySquared = this.velocity.squere();  
  let A = Math.PI * this.rocketRadius * this.rocketRadius;  
  let rho = this.air_rho();  
  let dx = -0.5 *  
    this.CD *  
    rho *  
    A *  
    velocitySquared *  
    this.velocity.normalize().getX() *  
    Number(Math.cos(this.rocketAngel * Math.PI / 180).toFixed(10))  
  let dy = -0.5 *  
    this.CD *  
    rho *  
    A *  
    velocitySquared *  
    this.velocity.normalize().getY() *  
    Number(Math.sin(this.rocketAngel * Math.PI / 180).toFixed(10))  
  
  return vector.create(  
    dx,  
    dy  
  );  
}
```





قوة الرفع:

```
// assume it's zero
// it will be more proffesional simulation if we can calculate it
// but in this case we wont do,
// because we assume that rocket maintains stability during its flight

function lift() {
  return vector.create(0, 0);
}
```



تحديد العوامل من قبل المستخدم:

▼ Rocket

Mass (kg)

100000

Fuel Mass

10000

Height

0.5

Radius

0.5

Nozzle Radius

1.5

▼ Head

CONE CD

HALF SPHERE CD

SHORT CYLINDER CD

▼ Forces

Temperature

30

▼ Engines

Rocketdyne F-1

Aerojet M-1

RD-170

Close Controls

كتلة الصاروخ

كتلة الوقود

ارتفاع الصاروخ

نصف قطر الصاروخ

نصف قطر الفوهة

رأس مدبب

رأس مكوّر

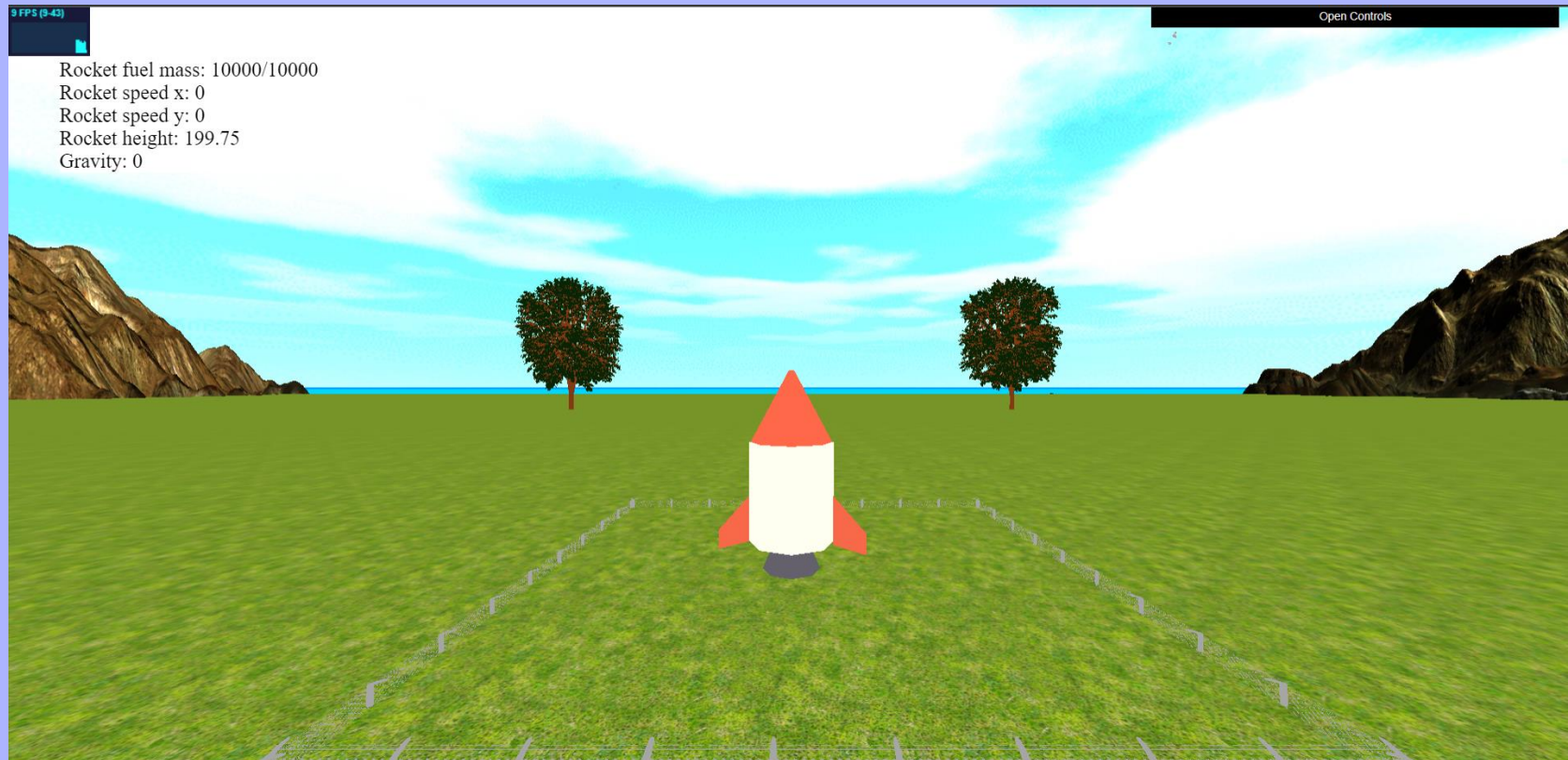
رأس مسطح

تغيير شكل الرأس

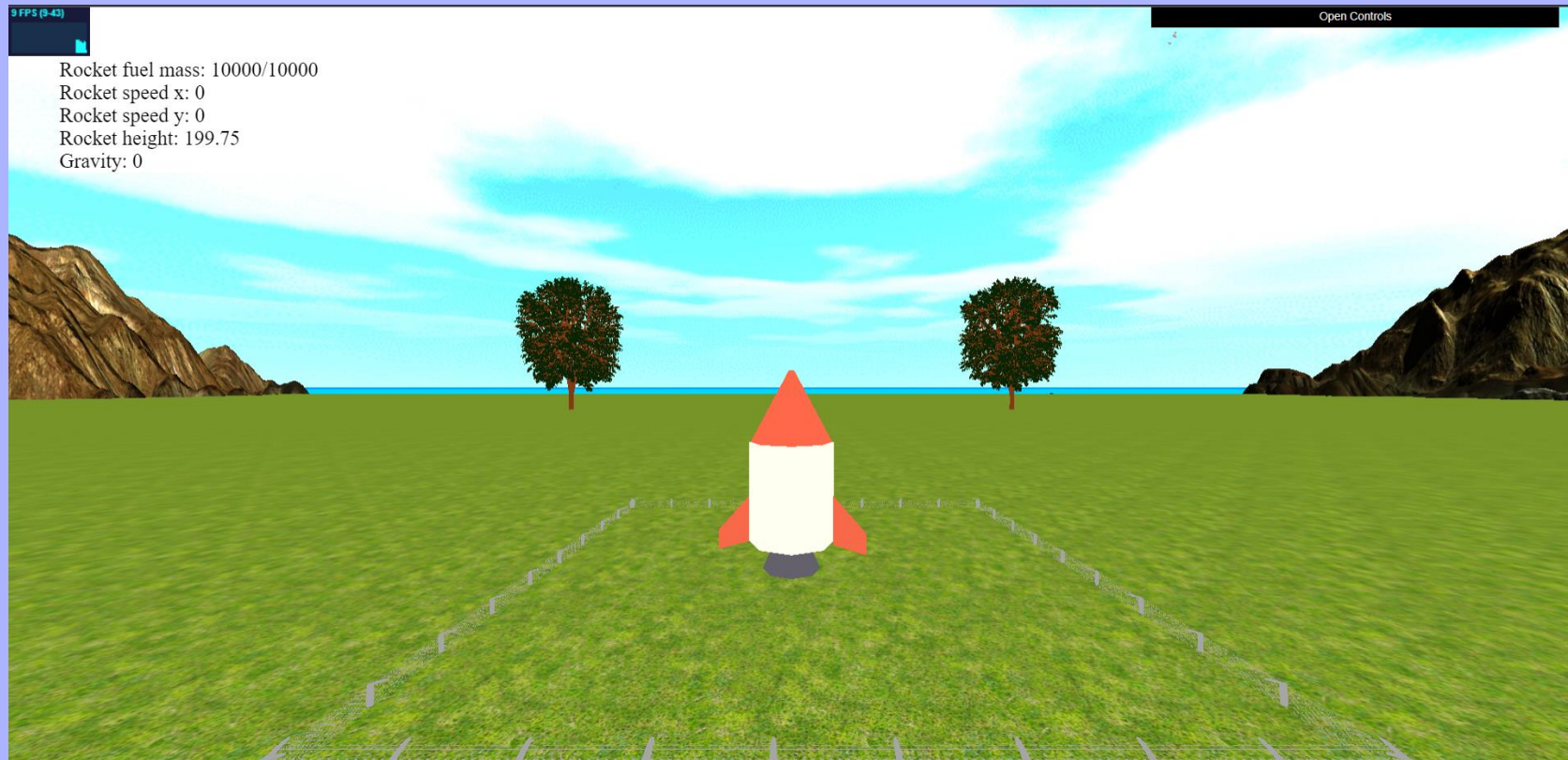
الحرارة

تغيير نوع المحرك

الإطلاق:



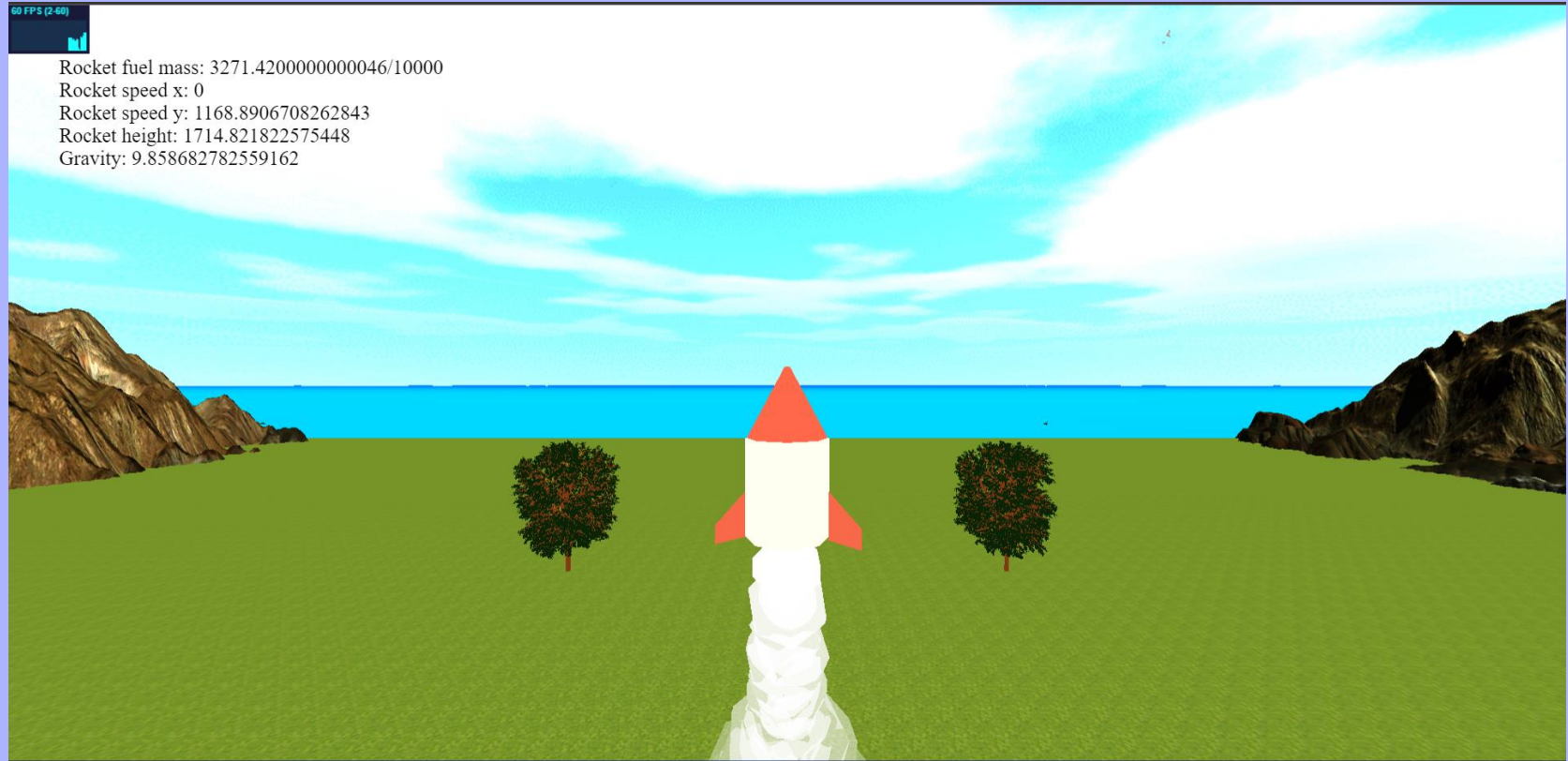
الإطلاق:



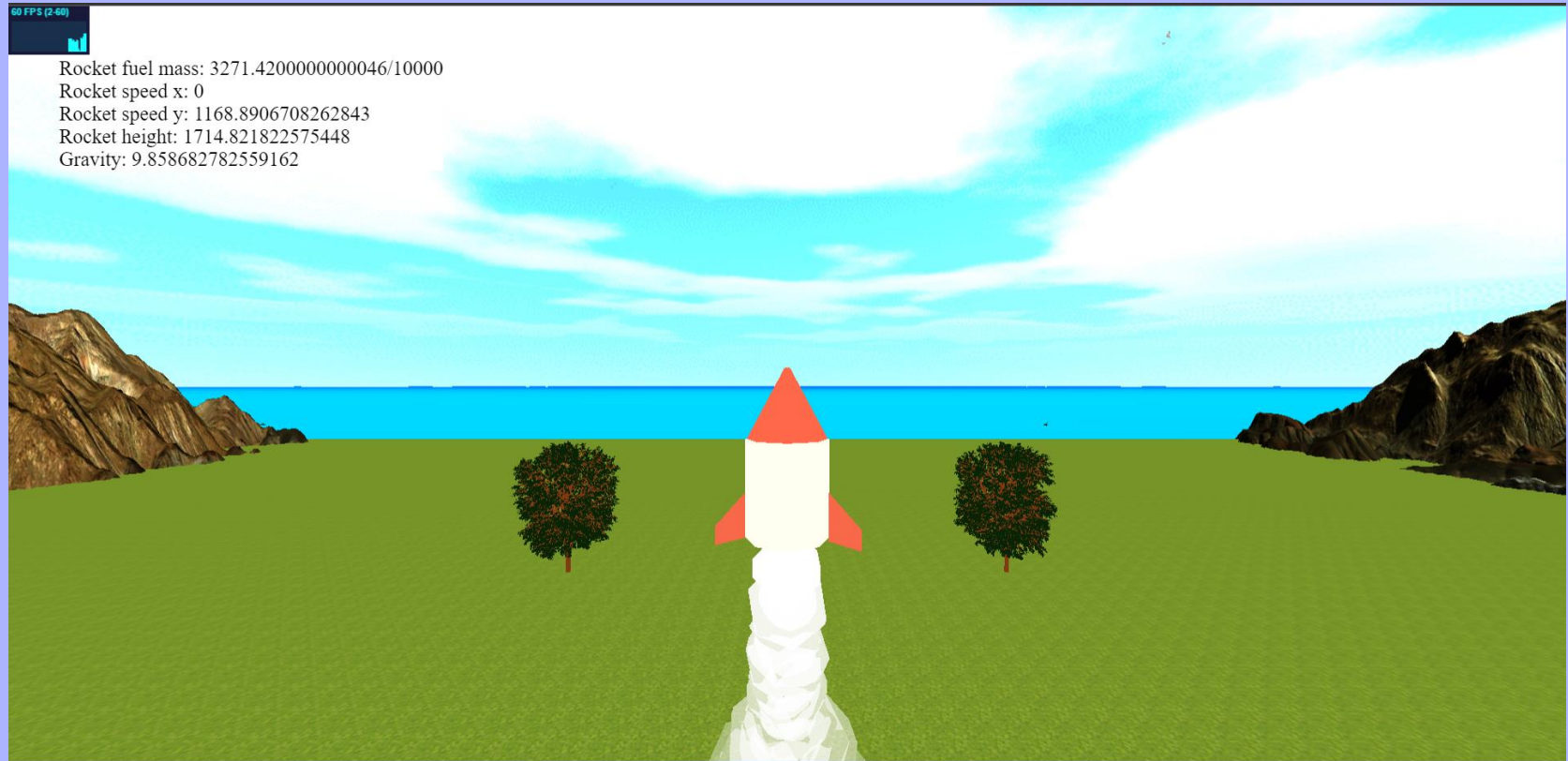
Press:



الإطلاق:



الإطلاق:



Press:

A

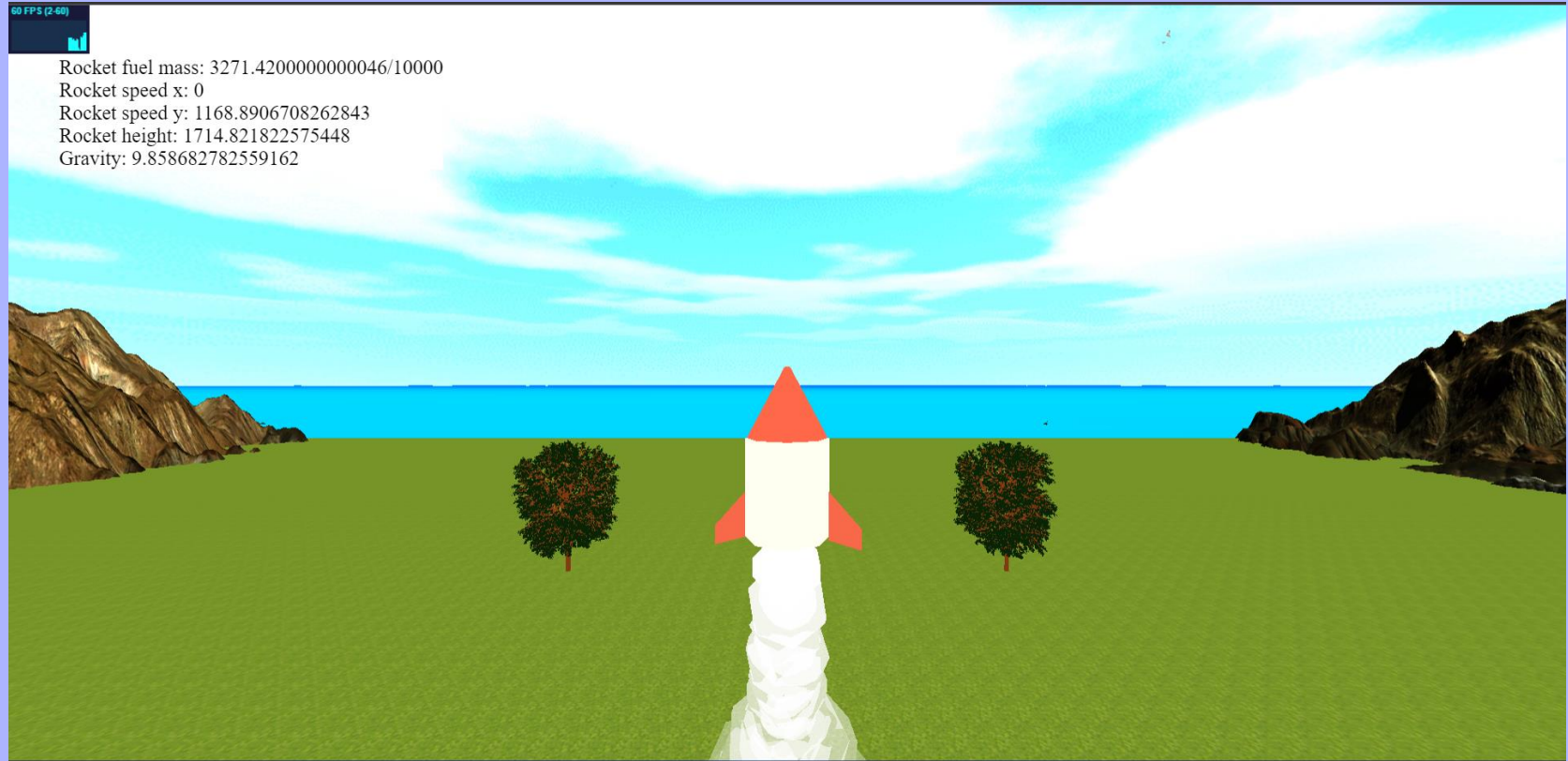
or

D

الإطلاق:



الإطلاق:



Press: ← or →

الإطلاق:





الإطلاق:

متابعة قيم المتحولات الأساسية

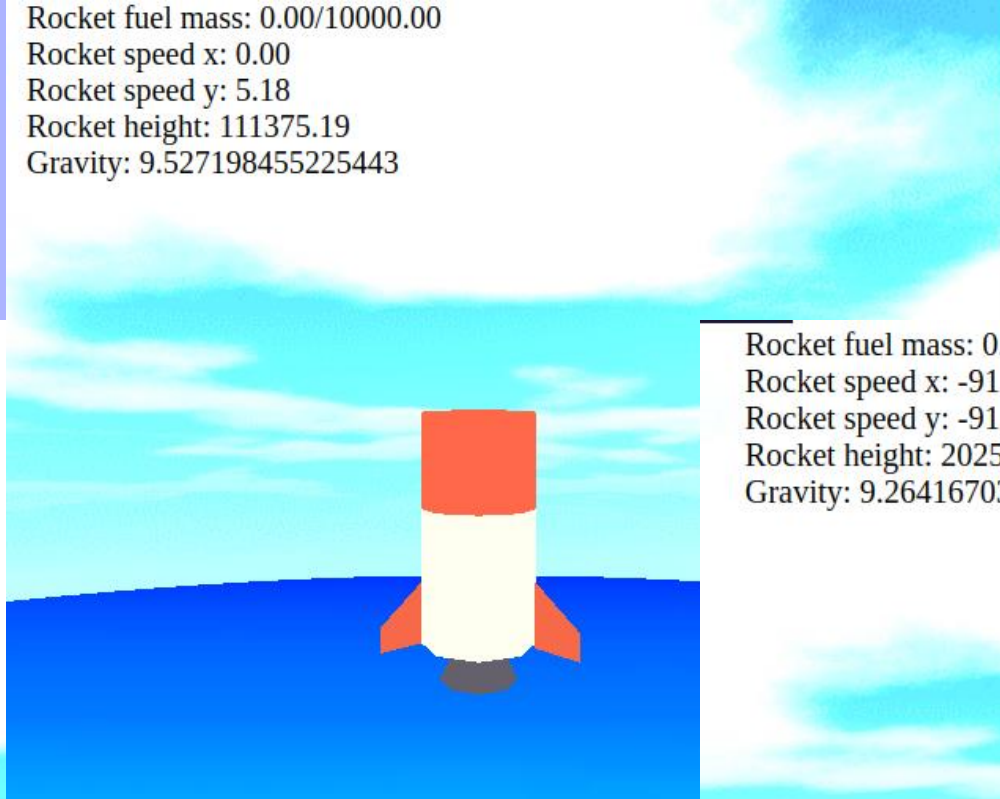
```
Rocket fuel mass: 5730.8320000000285/10000  
Rocket speed x: 0  
Rocket speed y: 733.3177088971323  
Rocket height: 806.2820248787461  
Gravity: 9.861488571023724
```





حالات مختلفة:

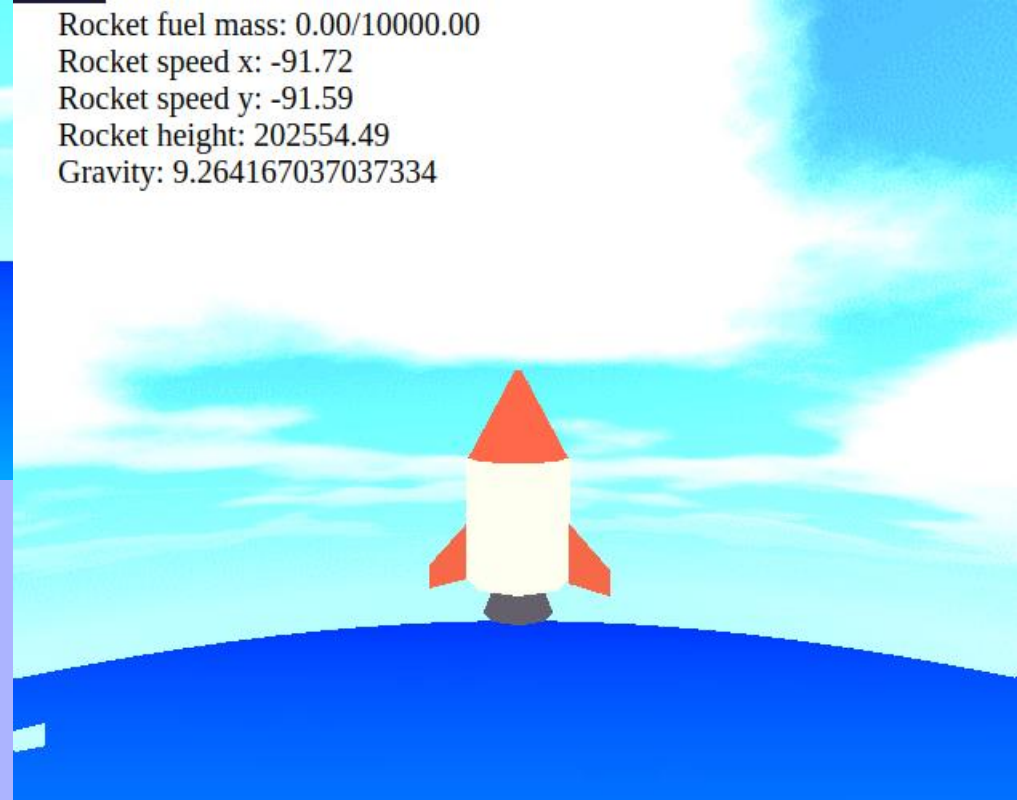
Rocket fuel mass: 0.00/10000.00
Rocket speed x: 0.00
Rocket speed y: 5.18
Rocket height: 111375.19
Gravity: 9.527198455225443



Rocket fuel mass: 0.00/10000.00
Rocket speed x: 0.00
Rocket speed y: 34.96
Rocket height: 218635.56
Gravity: 9.218911997127455



Rocket fuel mass: 0.00/10000.00
Rocket speed x: -91.72
Rocket speed y: -91.59
Rocket height: 202554.49
Gravity: 9.264167037037334

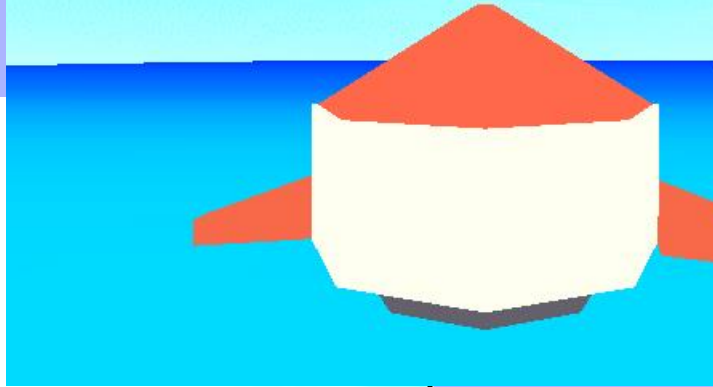




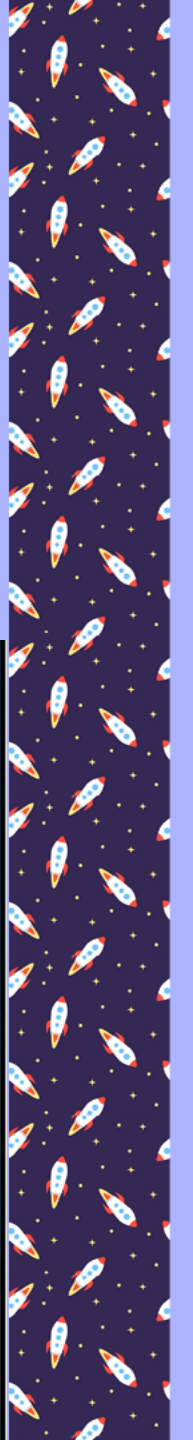
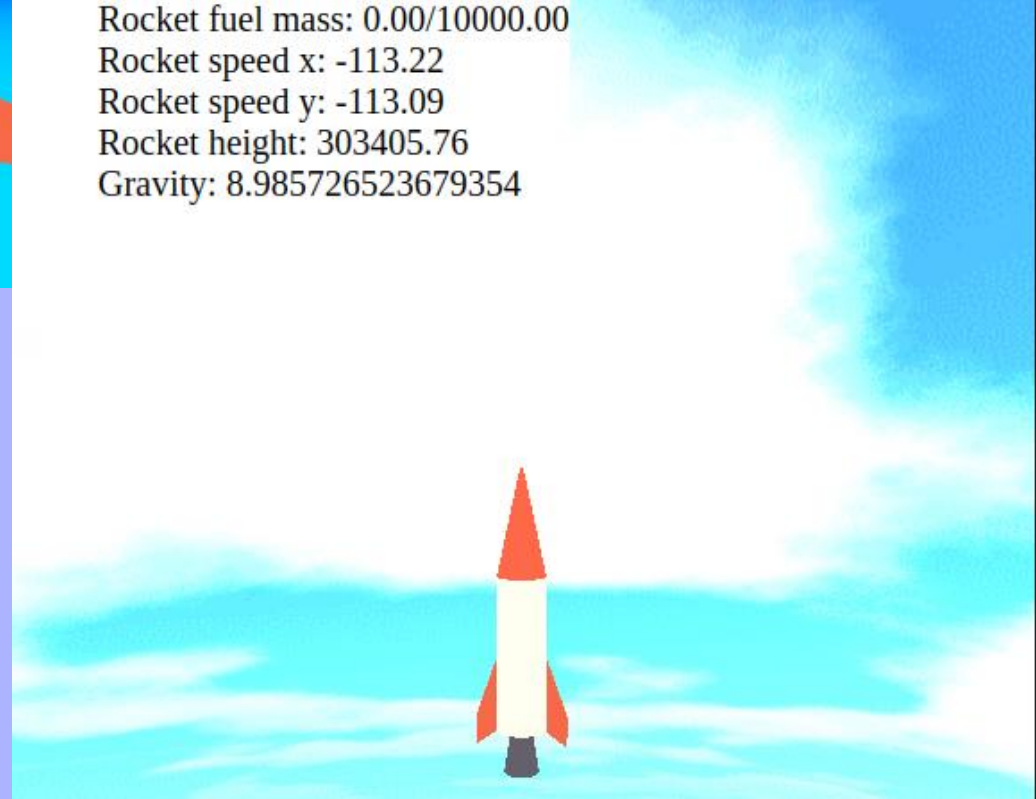
Rocket fuel mass: 0.00/10000.00
Rocket speed x: 0.00
Rocket speed y: 54.06
Rocket height: 23870.30
Gravity: 9.790306161196378

حالات مختلفة:

Rocket fuel mass: 0.00/10000.00
Rocket speed x: 0.00
Rocket speed y: 189.39
Rocket height: 202370.89
Gravity: 9.26469282152128

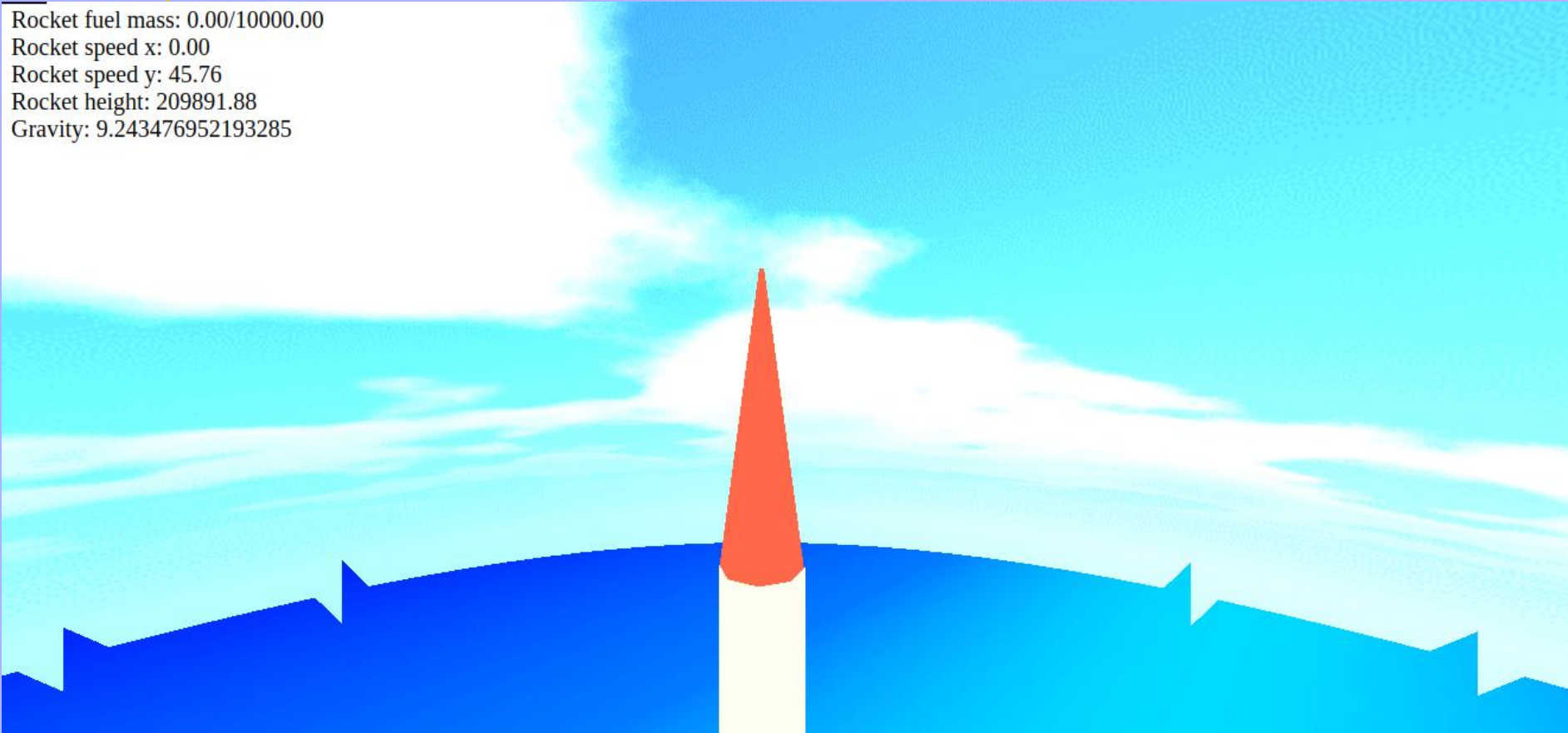


Rocket fuel mass: 0.00/10000.00
Rocket speed x: -113.22
Rocket speed y: -113.09
Rocket height: 303405.76
Gravity: 8.985726523679354



حالات مختلفة:

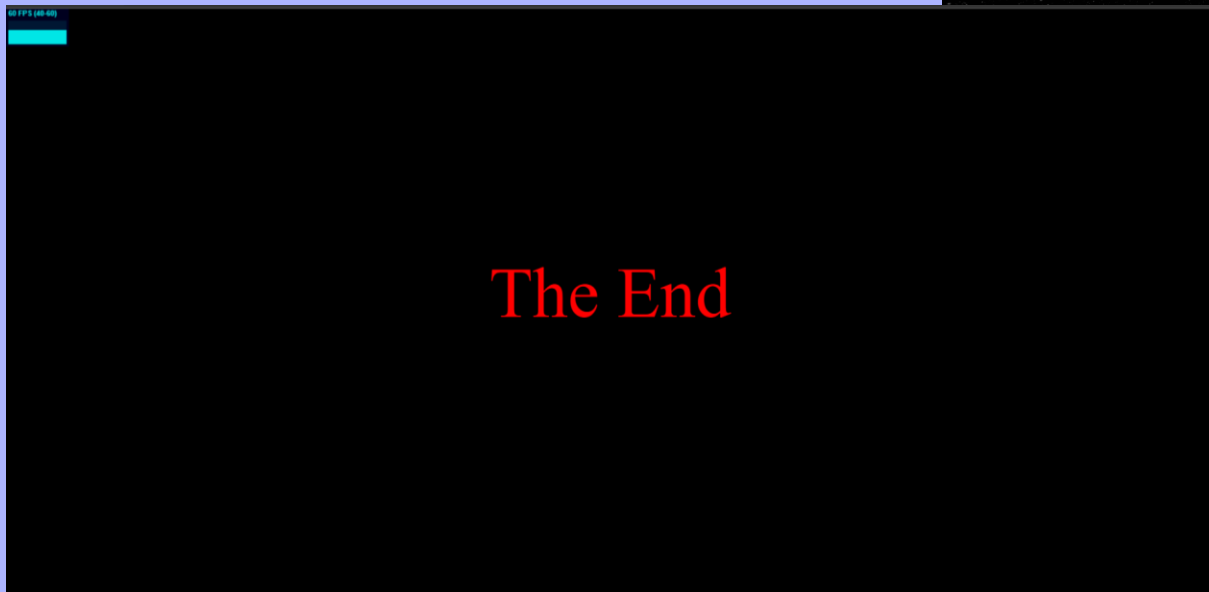
Rocket fuel mass: 0.00/10000.00
Rocket speed x: 0.00
Rocket speed y: 45.76
Rocket height: 209891.88
Gravity: 9.243476952193285



الوصول إلى الفضاء:



نهاية المحاكاة:





عمل الطلاب:

- حذيفة محمد مهدي زيادة
- محمد عقيل سعيد راعي
- نزار فايز أبو هايله
- ورد وديد الشاهين



The End

Thanks for being here

